MATTHEW WNUK

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CERTIFICATES AND PROFESSIONAL COURSES

NVIDIA-Certified: Generative AI Multimodal **December 2024**
NVIDIA-Certified: Generative AI LLMs **December 2024**
MIT professional education **July 2019**

- Modeling and Optimization for Machine Learning
- Designing Efficient Deep Learning Systems

EDUCATION AND HONORS

MSEE - University of California San Diego **June 2018**
BSEE - San Diego State University **December 2015**

Eta Kappa Nu (HKN) electrical engineering scholastic honor society – President
Tau Beta Pi (TBP) interdisciplinary engineering honor society

WORK HISTORY

Staff Embodied AI Engineer – Qualcomm — April 2026 to Present

- Embodied AI: Foundation models for humanoid and generalized robotics
  - Vision-Language-Action (VLA) policies and World-Action Models (WAM) for dexterous manipulation and whole-body control
  - Architecture design for in-house WAM foundation model: long-horizon task execution, failure detection and recovery, cross-embodiment adaptation
  - Hierarchical task conditioning – on-device VLM produces a latent subtask decomposition used to condition the action model
  - Gated DeltaNet selected over test-time training after hardware feasibility analysis
- Edge deployment and optimization – Qualcomm IQ-10
  - Latency and quantization analysis of world-action models (LingBot-VA, DreamZero, Fast-WAM, DiT4DiT, Cosmos Policy)
  - Per-stage roofline decomposition, quantization targets, KV-cache strategy, and ONNX export path
  - Diffusion transformer policy running on IQ-10 silicon at ~10 Hz
- Evaluation and data
  - Reproduction of published LIBERO benchmark results from released checkpoints to validate baselines
  - Design of large scale teleoperation data collection strategy

Senior Software Engineer (ML / Perception) – Symbotic — January 2023 to April 2026

- Tech Lead: CaseID
  - Project development for large scale production system from ideation, through literature review, model development and experimentation, through production deployment and monitoring.
  - Image embedding with Vision Transformers
  - Model training and development on Azure
  - Containerization, memory management, and deployment of onnx model in resource limited embedded ROS environment on NVIDIA Jetson
  - Design and development of deep learning pipeline using technologies such as OpenCV, PyTorch, pandas

Senior Machine Learning Engineer – Modern Intelligence **Aug 2022 to Nov 2022**

- Development of custom multi-modal reidentification system for maritime awareness

Staff Machine Learning Engineer – Sony Electronics **February 2020 to Aug 2022**

- Senior MLE: Machine Learning for Retail
  - Customization of YoloV5 architecture for embedded product recognition
  - Quantization and deployment of tflite on embedded camera
- Tech Lead: Visual Speech Recognition
  - Design and development of deep learning sequence to sequence pipeline for VSR using technologies such as OpenCV, Pytorch, Tensorflow, pandas, scikit learn
  - NLP for language understanding
  - Design of custom spatiotemporal CNN (3D) architectures in TensorFlow 2.x
  - Large-scale in-house data curation and preprocessing chain development
  - 100+ terabytes of data managed with Postgres SQL

Machine Learning Engineer – SPAWAR **January 2016 to February 2020**

- Tech Lead: Machine Learning for Self Defense
  - Random forest classifier using RF parametrics
  - Data cleaning and relational data base design (MySQL)
- Lead MLE: Radio Frequency Machine Learning Systems
  - Variational Autoencoders for HW fingerprinting and anomaly detection
  - DB-Scan (density-based clustering) in latent space
  - Manifold learning (t-sne embedding) for visualization of latent space
  - Large scale data curation for DARPA's RFML competition
  - Use of GAN for adversarial machine learning (early Generative AI)
  - Sci-kit Learn (sklearn) - random forest classifier using RF parametrics
- Trainer – Machine Learning
  - Topics include deep learning, linear regression, logistic regression, Naïve Bayes, Principal Component Analysis (PCA), unsupervised Learning, gradient descent, normalization techniques, Boosting and Bagging (ensemble methods), objective functions, and data augmentation

Center of Interdisciplinary Science – UCSD Research Lab **January 2018 to June 2018**

- Python Software development for Intel Aero Embedded Linux Environment
- SLAM – lidar based, occupancy grid.
- Particle filter for state estimation
- low level peripherals such as IMU & Magnetometer via SPI and I2C
- 9DOF Sensor Fusion for quaternion orientation via Madgwich Filter

Intern – NASA Langley Research Center **May 2015 to August 2015**

- Redesign of optical receiver for advanced lidar system
- Circuit design using RF design principals; PCB Layout in Altium Design Suite

AUVSI Robosub Competition **June 2014 to December 2015**

- First Place Team 2015 AUVSI International robotic submarine competition
- Designed and Built Custom Sonar System for Signal Localization and navigation
- Analog signal conditioning; coded 16-bit DSP for direction of arrival algorithm in embedded C

RF Engineer Intern – Northrop Grumman **June 2014 to January 2015**

- Software Development for Automation; macros for HFSS written in python
- Matlab scripting for development of database tools

Sonar Technician – United States Navy **July 2004 to August 2013**